

Jonas Conneryd

Department of Computer Science
Lund University
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Date of birth: July 31st, 1997
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Education

Lund University

Ph.D. in Theoretical Computer Science
Advisor: Prof. Tatyana Turova

Lund, Sweden
Fall 2026 (expected)

Lic. Eng. in Computer Science
Thesis: *On the Average-Case Proof Complexity of Graph Coloring*
Opponent: Prof. Nutan Limaye, ITU Copenhagen

Thesis defended May 2025

KTH Royal Institute of Technology

M.Sc. in Mathematics (joint with Stockholm University)
Thesis: *Geometric Bounds for Steklov Eigenvalues on Graphs* (awarded Mittag-Leffler Prize)

Stockholm, Sweden
June 2021

B.Sc. in Engineering Physics
Thesis: *Explicit Symplectic Integrators for Non-Separable Hamiltonians in Molecular Dynamics*

June 2019

Research Interests

Computational complexity theory, particularly proof complexity with an algebraic flavor and the complexity of (random) constraint satisfaction problems.

Publications

- [1] Jonas Conneryd, Yassine Ghannane, Jakob Nordström, Shuo Pang, Kilian Risse, and Dmitry Sokolov. *On the Power of Polynomial Calculus over Non-Boolean Domains*. Submitted. 2026.
- [2] Jonas Conneryd, Yassine Ghannane, and Shuo Pang. *Lower Bounds for CSP Hierarchies Through Ideal Reduction*. SODA 2026.
- [3] Jonas Conneryd, Susanna F. de Rezende, Jakob Nordström, Shuo Pang, and Kilian Risse. *Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz*. FOCS 2023.
- [4] Jonas Conneryd. *Geometric Bounds for Steklov Eigenvalues on Graphs*. M.Sc. thesis. Stockholm University, 2021.

Talks

ACM-SIAM Symposium on Discrete Algorithms (SODA) <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Vancouver, BC, Canada Winter 2026
KTH Royal Institute of Technology <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Stockholm, Sweden Fall 2025
Oxford Proof Complexity Workshop <i>Lower Bounds for CSP Hierarchies Through Ideal Reduction</i>	Oxford, United Kingdom Summer 2025
Proof Complexity and Beyond, Oberwolfach Workshop 2413 <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Oberwolfach, Germany Spring 2024
Algorithmic Research Cooperation around Øresund (ARCO) <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Odense, Denmark Fall 2023
IEEE Symposium on Foundations of Computer Science (FOCS) <i>Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz</i>	Santa Cruz, CA, USA Fall 2023

Research Visits and Workshops

Swiss Winter School on Theoretical Computer Science Visiting researcher for a winter school jointly organized by EPFL and ETH Zürich .	Zinal, Switzerland Winter 2026
KTH Royal Institute of Technology Month-long visit to the theory group.	Stockholm, Sweden Fall 2025
University of Oxford Visiting researcher for the workshop <i>Proof Complexity</i> in 2024 and 2025 .	Oxford, United Kingdom Summers 2024, 2025
Centre International de Rencontres Mathématiques Visiting researcher for the <i>Complexity as a Kaleidoscope</i> research school.	Marseille, France Spring 2025
DIMACS, Rutgers University Visiting researcher for the <i>Frontiers in Complexity Workshop</i> .	New Brunswick, NJ, USA Summer 2024
McGill University Week-long visit to the group of Asst. Prof. Robert Robere .	Montréal, Canada Summer 2024
Mathematisches Forschungsinstitut Oberwolfach Visiting researcher for the workshop <i>Proof Complexity and Beyond</i> .	Oberwolfach, Germany Spring 2024

Chalmers University of Technology

Visiting researcher for the *Workshop on Algebra and Computation*.

Simons Institute for the Theory of Computing at UC Berkeley

2-month visiting graduate student for the program *Satisfiability: Extended Reunion*.

Schloss Dagstuhl

Visiting researcher for the workshop *Theory and Practice of SAT and Combinatorial Solving*.

Gothenburg, Sweden

Summer 2023

Berkeley, CA, USA

Spring 2023

Dagstuhl, Germany

Fall 2022

Honors and Scholarships

- *WASP International Postdoctoral Scholarship (ca 100 000 EUR)*; 2026
- *WASP travel grant for one-month visit to McGill University (40 000 SEK)*; 2026
- *Oberwolfach Leibniz Graduate Student*; Mathematisches Forschungsinstitut Oberwolfach, 2024
- *WASP travel grant for two-month visit to Simons Institute at UC Berkeley (80 000 SEK)*; 2023
- *Accepted as WASP affiliated student*; 2021
- *Mittag-Leffler Prize for outstanding M.Sc. theses in mathematics*; Stockholm University, 2021
- *Ingenjör Ernst Johnson Scholarship for outstanding academic achievements*; KTH, 2020, 2021
- *Henrik Göransson Sandviken Scholarship for outstanding academic achievements*; KTH, 2018
- *University Merit Scholarship for outstanding academic achievements*; KTH, 2018, 2019, 2020, 2021

Service

Reviewed papers for the following venues:

- Conferences: FOCS '22; CCC '22; CCC '23; SAT '23; SAT '24; ICALP '24 (x2); AAI '23; AAI '24; SOSA '25; STOC '25; FOCS '26
- Journals: ACM Transactions on Computational Logic; Theoretical Computer Science; SIAM Journal on Discrete Mathematics

Teaching Experience

Lund University

- EDAA40/75 Discrete Structures in Computer Science, 2022, 2024, 2025, 2026
- EDAN55 Advanced Algorithms, 2023, 2024
- EDAN01 Constraint Programming, 2022, 2023

KTH Royal Institute of Technology

- SF1661 Perspectives on Mathematics, 2018
- SF1624 Algebra and Geometry, 2017

Work Experience

AP3 Third National Swedish Pension Fund

ILS Intern

Quantitative analysis of insurance-linked securities (ILS).

Stockholm, Sweden

2019–2021

Technical Skills

Programming: Python, Go, $\text{\LaTeX}$, Julia, MATLAB

Software: RMS Miu